

R²M: Real-Aware Residual Model Merging for Robust and Generalizable Deepfake Detection

Anonymous ECCV 2026 Submission

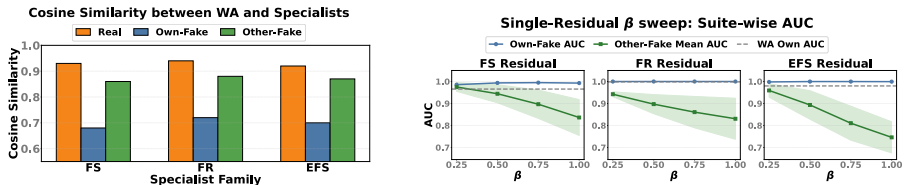
Paper ID #9145

Abstract. Deepfake generators evolve rapidly, making exhaustive data collection and repeated retraining impractical. Unlike generic multi-task settings, deepfake specialists share a common binary objective (Real vs. Fake) and mainly differ in generator-specific artifacts. However, naive parameter arithmetic can induce unintended decision-boundary shifts, causing unstable ranking behavior and degraded AUC under domain shift. We propose R²M, a training-free merging framework that decomposes task vectors into a shared component and generator-specific residuals, linking parameter-space updates to logit-space behavior. Offline spectral construction identifies shared and residual subspaces, and on-line routing selects residuals via first-order gradient-residual alignment, predicting per-sample logit updates under linearization. R²M is both efficient and interpretable: expensive computations are performed offline, while online inference requires only a single forward-backward pass with lightweight inner-product routing. The same formulation enables diagnostic analysis through margin and routing statistics. Experiments demonstrate strong performance in both in-domain and unseen settings while revealing measurable failure modes under real-domain mismatch, highlighting R²M as an interpretable and scalable approach to training-free deepfake model merging.

1 Introduction

Deepfakes have evolved from simple face splicing to photorealistic synthesis driven by modern generative models [5, 18, 32, 40]. Recent systems preserve identity while enabling precise control over lip movements and facial expressions, and high-quality content can now be generated through simple prompts on widely accessible generative platforms and APIs [27, 29]. While these advances democratize content creation, they amplify the spread of harmful and deceptive media, including financial fraud, copyright infringement, and political disinformation. These risks underscore the urgent need for robust deepfake detection methods. Because deepfake generation algorithms evolve rapidly, exhaustively collecting per-generator data and retraining a unified detector is fundamentally unsustainable. Even with sufficient data, jointly training on heterogeneous forgeries introduces gradient interference [36, 47]. Conversely, maintaining a dedicated specialist for each generator incurs substantial operational overhead [35, 44].

To address these challenges, we adopt a *model merging* paradigm for deepfake detection. Concretely, we first fine-tune specialist detectors on generator-



(a) Weight averaging preserves Real while suppressing Own-Fake alignment.

(b) Residual gains are selective and conflicting.

Fig. 1: Empirical observations motivating selective specialization in deepfake merging. (a) Weight averaging (WA) preserves shared Real structure while attenuating generator-specific fake cues. Cosine similarity between WA and each specialist in the pre-logit space remains high on *Real*, drops on *Own-Fake*, and is relatively higher on *Other-Fake*. (b) Injecting a single specialist residual improves performance on its corresponding forgery family over WA (dashed), but degrades others, exposing an inherent trade-off under unconditional merging. Solid curves show AUC of $\theta_{\text{WA}} + \beta(\theta_i - \theta_{\text{WA}})$ as β varies. Dashed horizontal lines indicate the WA baseline AUC on each dataset. The green shaded band shows the spread of AUC across non-target forgery families (from the lowest to the highest).

specific data and subsequently merge them directly in parameter space to obtain a single unified model. This strategy enables rapid incorporation of new generators without revisiting previous training data or performing costly joint retraining [14, 15, 41]. To the best of our knowledge, this is the first systematic investigation of model merging for deepfake detection.

Observation 1. To validate this design choice, we analyze how a weight-averaged (WA) model [15] relates to its specialists. We construct three specialists corresponding to distinct forgery families in DF40 [44]—FaceSwap (FS), FaceReenactment (FR), and Entire Face Synthesis (EFS)—all finetuned from the same pretrained backbone. A WA model is obtained by averaging their parameters, without additional retraining. We measure feature similarity between specialists and the WA model across Real, Own-Fake (generated by the specialist’s corresponding family), and Other-Fake images (generated by the remaining families). Interestingly, a consistent pattern emerges across all families. On Real images, specialists exhibit high similarity to the WA model, indicating that weight averaging preserves the shared Real representation. In contrast, similarity drops markedly on Own-Fake images, revealing that generator-specific signals—strongly encoded in individual specialists—are attenuated in the WA model. In contrast, similarity on Other-Fake images remains substantially higher than on Own-Fake images, as they lack the specialist’s dominant fake cues. As illustrated in Fig. 1(a), this behavior implies that WA preferentially retains the shared Real structure while suppressing generator-specific Fake residuals. This indicates that the components suppressed by WA encode generator-specific signals discarded during averaging.

Observation 2. To understand the role of parameters discarded by weight averaging, let θ_i denote the parameters of specialist i and θ_{WA} the weight-averaged model. We define the residual as $r_i = \theta_i - \theta_{\text{WA}}$ and construct interpolated models $\theta(\beta) = \theta_{\text{WA}} + \beta r_i$, where $\beta \in \mathbb{R}$ controls the residual strength. We

evaluate these models across the three forgery families. As shown in Fig. 1(b), increasing β consistently improves performance on the corresponding family relative to WA (dashed), while degrading performance on the others. Although $\beta = 1$ recovers the original specialist, even moderate β values exhibit pronounced trade-offs. This behavior indicates that residuals encode family-specific gains that conflict under unconditional merging. Consequently, no single global parameter setting can simultaneously realize all residual benefits, suggesting that specialization should be applied selectively rather than uniformly.

Proposed R²M. Building on these observations, we tailor merging to the structural properties of deepfake detection. Unlike conventional multi-task settings with disjoint label spaces, all specialists here operate on the same binary label space (Real vs. Fake). In practice, Real cues are largely stable and shared across datasets, whereas Fake cues are generator-specific and heterogeneous. Observations 1 and 2 therefore suggest that Real-aligned directions should be preserved globally, while Fake-specific directions should be activated conditionally to prevent cross-family interference. Accordingly, an appropriate merging strategy should: (i) preserve the shared Real structure, (ii) retain complementary generator-specific Fake knowledge, and (iii) apply Fake-specific gains selectively based on the input. We realize this via *Real-aware Residual Model Merging* (R²M). We first extract a shared Real-aligned component by applying low-rank factorization (*e.g.* SVD) to the specialists’ task vectors, interpreting the dominant directions as a core Real subspace. Each specialist update is then decomposed into a Real-aligned component and a residual encoding generator-specific Fake evidence. To enable efficient conditional specialization, we derive an input-conditional routing rule via a first-order margin approximation. As a result, R²M selects residuals through a single-gradient routing mechanism, eliminating multiple forward/backward evaluations of specialist models while preserving generator-specific discrimination.

Contributions. Our contributions are summarized as follows: (1) We present the first systematic study of model merging for deepfake detection, identifying a structural asymmetry: Real representations are largely shared across specialists, whereas Fake cues are generator-specific and conflict under unconditional merging. (2) We provide a geometric interpretation linking parameter-space factorization to feature-space structure, explaining why weight averaging preserves Real alignment and how R²M decouples Real and Fake effects. (3) We propose an efficient input-conditional routing mechanism derived from a first-order margin approximation, enabling selective residual application with a single backward pass. (4) Extensive experiments on DF40 and cross-dataset benchmarks demonstrate improved generalization and robustness to unseen forgery families.

2 Related works

Deepfake detection. Deepfake technology has advanced rapidly over the past decade. Early methods focused on localized facial manipulations [4, 19, 23], while GAN-based generators substantially improved realism [8, 31, 39], followed by diffusion models that further enhanced fidelity and controllability [32]. Beyond face replacement, modern talking-head and reenactment systems preserve identity

while enabling natural control over expressions, speech, and head motion [12, 17, 25, 26]. Commercial platforms such as Veo, Kling, and Wan have further lowered the barrier to high-quality synthetic video generation. While these advances support creative applications, they also introduce significant privacy and safety risks. On the detection side, prior work spans blending-based augmentation [20, 22, 35], frequency-domain analysis [16, 38, 48], and generalization to *unseen* forgeries [7, 10, 45]. Representative approaches include self-blending for pseudo-fake generation [35], shared forgery feature extraction [45], high-frequency modeling [38], and temporally aware curriculum learning [22]. Adaptor-based designs improve efficiency [10], and vision–language signals have also been explored [37]. More recently, [43] employ an SVD-based decomposition within a vision foundation model to separate pretrained semantic structure from forgery-specific cues, freezing principal components while adapting the orthogonal residual subspace.

Model merging. Model merging combines task-specific experts into a single model by operating directly in parameter space, typically without additional training. A basic strategy is weight averaging (WA) [15], which averages parameters across experts. In large-scale fine-tuning, WA underpins “model soups” and often improves accuracy and robustness without increasing inference cost [41]. Beyond uniform averaging, task arithmetic represents each task as a task vector—the difference between fine-tuned and pretrained weights—and modifies model behavior through vector addition or subtraction; combining multiple task vectors enables multi-task capability [14]. To mitigate interference, TIES-Merging trims small updates, resolves sign conflicts, and merges only sign-consistent parameters, improving multitask performance across modalities [42]. More recently, WA has been revisited from the task-vector perspective. CART centers task vectors around the weight average and applies low-rank approximation to reduce cross-task interference [6]. Orthogonal approaches such as *Fisher-weighted averaging* incorporate local curvature information during merging [24]. Recent work also explores combining model merging with test-time adaptation, where task coefficients or model parameters are adjusted during inference using gradient-based objectives [46].

Mixture-of-Experts routing. Mixture-of-Experts (MoE) models scale capacity by activating only a subset of expert modules per input using a learned router [34], with later variants simplifying routing and improving training stability [11]. Subsequent work improves routing stability and load balancing, including expert-choice routing variants [49]. Recent efforts also explore converting dense checkpoints into MoE-style models for efficient deployment [50].

Positioning of Our Work. To the best of our knowledge, model merging has not been systematically studied for deepfake detection. Existing merging approaches are largely task-agnostic, treating task updates symmetrically. In contrast, deepfake detection exhibits a structural asymmetry: Real representations are largely shared across specialists, whereas Fake cues are generator-specific. R²M exploits this property by preserving a shared Real component while selectively activating generator-specific residuals, yielding a unified detector that remains composable as new forgery families are introduced.

Our approach is complementary to training-time generalization methods, which aim to train a single detector with stronger cross-generator robustness. Instead, we address the deployment problem of merging multiple independently fine-tuned specialists directly in parameter space without retraining.

A closely related line of work studies subspace factorization for AI-generated image detection. [43] propose an orthogonal SVD-based decomposition that separates pretrained semantic structure from forgery-specific cues during training. While both approaches share the idea of separating shared and forgery-specific information, the problem setting differs fundamentally: their method improves generalization through additional training on a single detector, whereas R²M targets training-free merging of multiple specialists via parameter-space spectral decomposition and input-conditional residual activation.

Finally, our conditional specialization is conceptually related to Mixture-of-Experts routing and test-time adaptation. Unlike MoE methods that learn routing networks and modify the architecture, R²M derives routing analytically from first-order logit dynamics to select compatible residual directions. Moreover, in contrast to test-time adaptation methods that update model parameters during inference, R²M performs no test-time optimization: gradients are used only as an analytic signal for routing, leaving model parameters unchanged.

3 Method

Let $f_\theta(x) \in \mathbb{R}^2$ denote the logits of a binary deepfake detector parameterized by θ for an input image sample x , corresponding to the classes Real and Fake. We define the classification margin as

$$z(\theta, x) = f_\theta^{\text{fake}}(x) - f_\theta^{\text{real}}(x), \quad (1)$$

so that $z(\theta, x) > 0$ indicates Fake and $z(\theta, x) < 0$ indicates Real. This margin formulation is natural for deepfake detection, which is typically evaluated using ranking-based metrics such as AUC. Let θ_0 denote a pretrained backbone parameter and $\theta_i = \theta_0 + \Delta_i$ denote a specialist parameter trained on forgery family i , where Δ_i is its task vector. We decompose each task vector as $\Delta_i = c + r_i$, where c represents a shared component and r_i captures family-specific residual updates. This decomposition exposes the shared and family-specific structure of task vectors, making (P1)–(P3) analyzable through explicit geometric constraints. We consider a merged model of the form $\theta^*(x) = \theta_0 + c + \beta r_{i(x)}$, where $i(x)$ denotes an input(x)-dependent routing rule selecting a residual component, and β control the contributions of the shared and residual components. For convenience, we define the shared model: $\theta_c := \theta_0 + c$. We then define the residual-induced margin shift as

$$\Delta z_i(x) = z(\theta_c + \beta r_i, x) - z(\theta_c, x), \quad (2)$$

which quantifies how the residual component r_i perturbs the classification margin relative to the shared model. Finally, let $\mathcal{D}$ denote the overall data distribution, which consists of Real and Fake samples. We denote by $\mathcal{D}_{\text{real}}$ the marginal distribution of Real samples, and by $\mathcal{D}_{\text{fake}}^{(i)}$ the conditional distribution of Fake samples from forgery family i . Then, a desirable merged model should satisfy the following properties:

(P1) Real invariance (residual robustness). $\mathbb{E}_{x \sim \mathcal{D}_{real}}[|\Delta z_i(x)|] \leq \epsilon$, for each residual r_i , where $\epsilon > 0$ is a small tolerance controlling Real robustness. This condition ensures that residual updates do not significantly perturb the margin on Real samples, preserving the shared Real decision structure and keeping residual effects fake-specific.

(P2) Fake selective gain (own-family amplification). For Fake samples from family i , $\mathbb{E}_{x \sim \mathcal{D}_{fake}^{(i)}}[\Delta z_i(x)] \geq m$, and for $j \neq i$, $\mathbb{E}_{x \sim \mathcal{D}_{fake}^{(i)}}[\Delta z_j(x)] \leq m'$, where $m > m' \geq 0$ are desired margin gains for own- and cross-family samples, respectively. This enforces that each residual selectively increases the margin for its corresponding forgery family, while exerting limited influence on others.

(P3) Local Stability (Cross-Task Calibration). Residual perturbations should remain small relative to the shared margin: $\mathbb{E}_{x \sim \mathcal{D}}\left[\frac{|\Delta z_i(x)|}{|z(\theta_c, x)| + \delta}\right] \leq \tau$, for small constants $\delta, \tau > 0$. This prevents residual updates from dominating the shared decision structure and preserves the local linear approximation.

Design Objective. Our goal is to design a decomposition and routing mechanism that approximately satisfies (P1)–(P3).

3.1 Real-Fake Structured Decomposition

We aim to construct a decomposition $\Delta_i = c + r_i$ that approximately satisfies (P1) real invariance and (P2) fake selective gain. We first characterize the underlying geometric conditions required for these properties, and then show that they naturally admit a spectral solution.

Optimal Residual Geometry under Real-Fake Constraints. For the shared model $\theta_c = \theta_0 + \alpha c$, the residual-induced margin shift $\Delta z_i(x)$ is defined in Eq. (2). Applying a first-order Taylor expansion, $f(\theta + \delta) \approx f(\theta) + \nabla_\theta f(\theta)^\top \delta$, we obtain

$$\Delta z_i(x) \approx \beta \nabla_\theta z(\theta_c, x)^\top r_i = \beta \langle g(x), r_i \rangle, \quad (3)$$

where $g(x) := \nabla_\theta z(\theta_c, x)$ denotes the gradient of the margin evaluated at the shared model. Under this linearization, the effect of a residual direction r_i is governed by its alignment with the gradient evaluated at the shared model θ_c . In other words, residual-induced margin shifts depend on how strongly r_i aligns with the local geometry of the margin at θ_c .

To analyze (P1) in a tractable manner, we consider the expected squared margin shift on Real samples. Minimizing the squared shift provides a smooth surrogate for controlling $|\Delta z_i(x)|$. Using the first-order approximation derived in Eq. (3), $\Delta z_i(x)^2 \approx \beta^2 \langle g(x), r_i \rangle^2 = \beta^2 r_i^\top g(x) g(x)^\top r_i$. Taking expectation over Real samples yields $\mathbb{E}_{x \sim \mathcal{D}_{real}}[\Delta z_i(x)^2] \approx \beta^2 r_i^\top \Sigma_R r_i$, where Σ_R is the real gradient covariance matrix, *i.e.* $\Sigma_R = \mathbb{E}_{x \sim \mathcal{D}_{real}}[g(x) g(x)^\top]$. Thus, Real sensitivity is governed by the quadratic form induced by Σ_R . Let the eigendecomposition be $\Sigma_R = U_R \Lambda_R U_R^\top$. Directions associated with large eigenvalues correspond to parameter perturbations that strongly affect the Real margin.

To satisfy (P1) under the squared-shift surrogate, we require each residual to have limited Real sensitivity:

$$r_i^\top \Sigma_R r_i \leq \epsilon_R, \quad \epsilon_R > 0. \quad (4)$$

A sufficient condition is to remove the dominant eigenspace of Σ_R , motivating the projection step introduced next.

For Fake samples from family i , $\mathbb{E}_{x \sim \mathcal{D}_{\text{fake}}^{(i)}} \Delta z_i(x) \approx \beta r_i^\top \mathbb{E}_{x \sim \mathcal{D}_{\text{fake}}^{(i)}} [g(x)]$. Let $\mu_{F,i} := \mathbb{E}_{x \sim \mathcal{D}_{\text{fake}}^{(i)}} [g(x)]$. Then **(P2)** promotes residual directions that increase $r_i^\top \mu_{F,i}$, *i.e.* directions aligned with the mean Fake gradient of family i . Combining this with the Real-invariance constraint in Eq. (4) yields the following constrained trade-off:

$$\max_{r_i} r_i^\top \mu_{F,i} \quad \text{s.t.} \quad r_i^\top \Sigma_R r_i \leq \epsilon_R, \quad (5)$$

which formalizes the requirement that residual directions should align with fake-specific gradients while avoiding dominant real-sensitive modes.

Spectral Interpretation. The constrained optimization in Eq. (5) admits a closed-form solution characterized by the spectrum of Σ_R .

Proposition 1 (Optimal Residual under Real–Fake Trade-off). *Let $\Sigma_R = U\Lambda U^\top$ be the eigendecomposition of the real gradient covariance matrix, and let $\mu_{F,i}$ denote the fake-gradient mean of family i . Consider the constrained optimization problem $\max_{r_i} r_i^\top \mu_{F,i}$ s.t. $r_i^\top \Sigma_R r_i \leq \epsilon_R$. An optimal solution is given by $r_i^* \propto U\Lambda^\dagger U^\top \mu_{F,i} = \sum_{k=1}^d \frac{u_k^\top \mu_{F,i}}{\lambda_k} u_k$, where λ_k and u_k denote the eigenvalues and eigenvectors of Σ_R , and $\Lambda^\dagger$ is the Moore–Penrose pseudoinverse of Λ .*

Implication. The optimal residual scales each fake-gradient component by the inverse of its corresponding Real-sensitivity eigenvalue. Consequently, directions associated with large λ_k (*i.e.* dominant Real-sensitive modes) are suppressed, while alignment with fake-specific gradients is retained. The proof is provided in Appendix A.2

While Proposition 1 characterizes the optimal direction for a single family r_i , practical merging requires a shared low-rank residual structure across families. We therefore seek a rank- k subspace $W \in \mathbb{R}^{d \times k}$ such that each residual r_i lies in $\text{span}(W)$. A natural surrogate for jointly preserving fake-specific alignment is the aggregated projection energy:

$$\sum_{i=1}^T \|W^\top \mu_{F,i}\|^2 = \text{Tr}(W^\top M_F W), \quad M_F := \sum_{i=1}^T \mu_{F,i} \mu_{F,i}^\top, \quad (6)$$

which measures how well the subspace W captures the family-wise Fake-gradient means. Using the trace formulation in Eq. (6), we next characterize the optimal low-rank subspace under the Real-sensitivity constraint.

Proposition 2 (Optimal Low-Rank Residual Subspace). *Let $\Sigma_R \succeq 0$ denote the real-gradient covariance matrix, and let M_F be defined in Eq. (6). To obtain a shared rank- k residual structure, we consider the constrained spectral problem: $\max_{W^\top W = I_k} \text{Tr}(W^\top M_F W)$ s.t. $\text{Tr}(W^\top \Sigma_R W) \leq \epsilon_R$. Then, any optimal solution W^* spans the top- k generalized eigenspace of (M_F, Σ_R) . Equivalently, W^* is obtained by spectral decomposition of the whitened matrix $\Sigma_R^{-1/2} M_F \Sigma_R^{-1/2}$.*

Implication. The optimal residual subspace arises from a generalized spectral trade-off between fake-specific alignment and Real robustness. Whitening by $\Sigma_R^{-1/2}$ attenuates directions that strongly influence Real margins, while M_F promotes directions shared across Fake families. Consequently, the learned subspace captures fake-discriminative variations that lie in the complement of dominant Real-sensitive modes. The proof is provided in Appendix [A.3](#).

Scale Ambiguity and Calibration. Proposition [2](#) determines the optimal residual *subspace*, but not its scale. Since the generalized spectral objective is scale-invariant, each residual direction is identifiable only up to task-wise scaling. Furthermore, Stage 1 removes dominant real-sensitive modes without fully equalizing variance across tasks. As a result, residual magnitudes may differ across branches, potentially biasing downstream selection mechanisms. This motivates an additional calibration step to approximately satisfy (P3), as described in the next section. Importantly, unlike (P1) and (P2), which impose structural constraints on residual geometry in parameter space, (P3) governs inference-time perturbation behavior and therefore lies beyond purely spectral characterization.

3.2 Spectrally-Motivated Practical Construction

Propositions [1](#) and [2](#) establish that the optimal residual structure is governed by a generalized spectral problem involving the Real-gradient covariance Σ_R and the Fake second-moment matrix M_F . In particular, the optimal residual subspace W^* corresponds to the top- k eigenspace of the whitened matrix $\Sigma_R^{-1/2} M_F \Sigma_R^{-1/2}$. Directly computing Σ_R and M_F is infeasible in high-dimensional parameter space. We therefore construct an efficient low-rank approximation that preserves the same spectral geometry while remaining tractable.

Conceptually, Proposition [2](#) reveals that our method performs PCA on Fake gradients after removing the geometry induced by Real gradients. The whitening step (*Stage 1*) suppresses Real-sensitive directions, and the subsequent spectral decomposition (*Stage 2*) extracts dominant Fake-structured components in this normalized space. In practice, this corresponds to projecting out dominant Real modes and then performing spectral analysis on the residual task vectors. Overall, the proposed two-stage procedure yields a truncated and sample-efficient approximation to the theoretically optimal generalized spectral solution.

Stage 1: (Approximate Real-Whitening via Subspace Removal). Proposition [1](#) indicates that optimal residual directions should attenuate the dominant eigendirections of Σ_R . Let $g(x) = \nabla_{\theta} z(\theta_c, x)$ denote the margin gradient evaluated at the shared model. Using a small set of Real diagnostic samples, we construct the gradient matrix $G_R = [g(x_1), \dots, g(x_n)]$, $x_j \sim \mathcal{D}_{\text{real}}$. Since $\Sigma_R \approx \frac{1}{n} G_R G_R^\top$, its principal eigenspace can be approximated via truncated SVD of G_R . Let U_R denote the top- r_0 left singular vectors. We suppress these real-sensitive modes by projecting each task vector Δ_i onto the orthogonal complement: $\Delta'_i = (I - U_R U_R^\top) \Delta_i$. This operation approximates whitening by suppressing the dominant eigenspace of Σ_R . Rather than explicitly rescaling by $1/\lambda_k$, we remove the principal Real-sensitive modes, which suffices to enforce (P1) in practice.

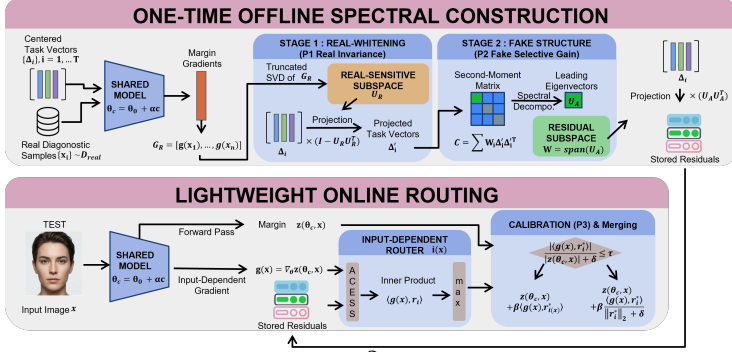


Fig. 2: Overall framework of the R²M The process is divided into two main phases: (Top) **One-time Offline Spectral Construction**: This phase identifies the optimal residual structure through two stages: *Stage 1 (Approximate Real-Whitening)* suppresses dominant Real-sensitive modes to ensure P1; *Stage 2 (Spectral Approximation)* extracts the leading Fake-structured components from projected task vectors to satisfy P2. (Bottom) **Lightweight Online Routing**: For a test sample x , a single backward pass gets the margin gradient $g(x)$, allowing the input-dependent router to select the most aligned residual $r^*(x)$ and approximate the merged margin z_{merged} .

Stage 2: (Spectral Approximation of Fake Second-Moment Structure). Proposition 2 shows that the optimal residual subspace is determined by the spectral structure of $M_F = \sum_i \mu_{F,i} \mu_{F,i}^\top$. In practice, the true Fake-gradient means $\mu_{F,i}$ are unknown and expensive to estimate precisely. Task vectors Δ_i , being optimized to minimize family-specific Fake losses, naturally tend to align with $\mu_{F,i}$. We therefore use the projected task vectors Δ'_i as proxies for Fake-gradient directions. To approximate M_F , we construct a weighted second-moment matrix in task space: $C = \sum_{i=1}^T w_i \Delta'_i \Delta'^T_i$, where weights w_i capture the degree of family-specific gradient alignment estimated from diagnostic samples. We then compute the top- r_A eigenvectors of C , denoted by U_A . The final subspace $W = \text{span}(U_A)$ serves as a low-rank surrogate for the leading eigenspace of $\Sigma_R^{-1/2} M_F \Sigma_R^{-1/2}$.

3.3 Practical Implementation

In Fig. 2, our framework consists of a *one-time offline spectral construction* followed by *lightweight online routing*. In the offline stage, we analyze gradients to estimate the dominant real-sensitive subspace using a small diagnostic set of real samples (24 images). After projecting this subspace out from centered task vectors, we construct a second-moment matrix whose leading eigenvectors define the residual subspace; for simplicity and interpretability, we use uniform weights ($w_i = 1$) and precompute specialist residual directions for inference.

At test time, a single backward pass computes the margin gradient, and routing scores are obtained via inner products with the stored residual directions to select the most aligned specialist. The merged margin is then approximated by a first-order update without additional forward passes. Since spectral operations are performed only once and inference requires only one gradient computation and T inner products, the method scales efficiently with minimal overhead.

Inference-time calibration further regulates branch scaling to keep residual perturbations within the local linear regime, approximately satisfying (P3).

Specifically, we use $\rho(x) = \frac{|\Delta z(x)|}{|z_c(x)| + \delta}$ as a linear-stability indicator and, when needed, replace $\Delta z = \beta \langle g, r_{i^*} \rangle$ with $\Delta z = \beta \frac{\langle g, r_{i^*} \rangle}{\|r_{i^*}\|_2 + \epsilon}$ to control branch scale. Implementation details are provided in Appendix A.4

4 Experiments

Benchmark and protocols (DF40). We evaluated on DF40 [44], a recent deepfake detection benchmark that implements 40 manipulation/generation methods across four *forgery categories*: face swapping (FS), face reenactment (FR), entire-face synthesis (EFS), and face editing (FE). We followed DF40 naming of data “domains” (e.g. FF++ [33] and Celeb-DF/CDF [21]). DF40 standardizes three evaluation protocols: (i) *Protocol 1 (cross-forgery, same domain)*: train and test within the same data domain while varying forgery methods; (ii) *Protocol 2 (cross-domain, same forgery)*: train and test on the same forgery category while changing the data domain; (iii) *Protocol 3 (unknown forgery & domain)*: train on seen forgeries/domains and test on *unseen* forgeries and domains to simulate open-set conditions. These protocol definitions, the four-category taxonomy (FS/FR/EFS/FE), and FF++/CDF domains were adopted from DF40.

Specialist Model Construction and Training Policy. We trained three specialists, one per forgery category: θ_{FS} , θ_{FR} , and θ_{EFS} , each trained on the union of methods within its category (8 methods per category; 24 total). We also trained a single *all-in-one* model on all 24 methods. All models use CLIP-L/14 [30] as the backbone with `pooler_output` embeddings and a binary linear head trained via cross-entropy. Since our goal is to study training-free model merging for diverse forgery types, we train all specialists under an identical architecture and training recipe to ensure a controlled comparison. All training followed the official DF40 splits, with no data from Protocols 2 or 3 used for training or tuning. Training details and the full method list are provided in Appendix B.1.

Evaluation metrics. We reported image-level area under the ROC curve (AUC) under all three DF40 protocols and compared merged models against specialists and the all-in-one baseline. All evaluations followed the DF40 train/test splits and protocol definitions. We compared against training-free merging baselines.

Model Merging Hyperparameter Selection. Hyperparameters were selected using Protocol 1 validation only and fixed across Protocols 2 and 3 to prevent cross-protocol leakage. For our method, the hyperparameters consist of two offline ranks and one online scaling factor. r_0 controls the dimension of the Real-sensitive subspace removed in Stage 1 (top- r_0 singular vectors U_R), r_A determines the rank of the residual subspace extracted in Stage 2 (top- r_A eigenvectors U_A of C), and β is the residual scaling coefficient applied at inference time. We searched $r_0 \in 0, 1, 2$ and $r_A \in 1, 2, 3$, and selected β from 0.6, 0.8, 1.0. For baseline merging methods, we followed their hyperparameter grids; formulations are detailed in Appendix B.2. To ensure fair comparison, we applied the same search ranges across all merging methods with the same averaged head. Experiments related to the classifier head are included in Appendix B.4

4.1 Main Results on DF40 (Protocols 1–3)

Tab. 1 summarizes per-category AUCs under Protocols 1 and 2, covering both in-domain (FF++) and cross-domain (Celeb-DF) settings.

Table 1: DF40 Protocols 1–2 Results (higher is better). Columns are category×domain + domain-wise means. Protocol 1: cross-forgery within FF++ (in-domain). Protocol 2: cross-domain transfer to Celeb-DF (CDF). Best results are shown in **bold**, second-best are underlined.

Method	Protocol1 - FF				Protocol2 - CDF			
	FS	FR	EFS	Mean	FS	FR	EFS	Mean
<i>Our DF40-compliant training</i>								
Specialist-FS	0.995	0.912	0.766	0.891	0.959	0.690	0.603	0.751
Specialist-FR	0.923	0.999	0.742	0.888	0.505	0.915	0.099	0.506
Specialist-EFS	0.676	0.814	0.999	0.830	0.618	0.621	0.989	0.742
All-in-one	<u>0.962</u>	<u>0.997</u>	<u>0.978</u>	0.979	<u>0.759</u>	<u>0.860</u>	0.366	0.662
<i>Training-free merging baselines</i>								
Weight Averaging	0.968	0.997	0.982	0.982	0.825	0.909	0.767	0.834
Task Arithmetic	0.956	0.995	0.965	0.972	0.741	0.888	0.926	0.852
TIES-Merging	0.959	0.993	0.961	0.971	0.819	0.898	0.891	0.869
CART	0.976	0.994	0.995	0.988	0.851	0.874	0.907	0.877
R²M (ours)	0.992	0.999	0.997	0.996	<u>0.829</u>	0.932	0.796	0.852

Table 2: DF40 Protocol 3 Results (higher is better). *Unseen datasets:* DeepFaceLab [23], HeyGen [1], Midjourney [2], WhichIsReal [3], StarGAN [8], StarGAN v2 [9], StyleCLIP [28], CollabDiff [13].

Method	DeepFace Lab	Hey Gen	Mid journey	WhichIs Real	Star Gan	StarGan v2	Style Clip	Collab Diff	Mean
<i>Our DF40-compliant training</i>									
Specialist-FS	0.892	0.573	0.334	0.364	0.887	0.640	0.608	0.596	0.612
Specialist-FR	0.812	0.546	0.603	0.228	0.697	0.433	0.053	0.141	0.439
Specialist-EFS	0.709	0.398	<u>0.561</u>	0.694	0.901	0.777	0.952	0.997	0.749
All-in-one	<u>0.884</u>	<u>0.561</u>	0.177	<u>0.512</u>	0.860	0.714	0.761	0.854	0.665
<i>Training-free merging baselines (same backbone)</i>									
WA	0.930	0.626	0.613	0.455	0.953	0.728	0.635	0.848	0.724
TA	0.887	0.541	0.695	0.255	0.891	0.579	0.504	0.622	0.622
TM	0.897	0.579	0.545	0.327	0.912	0.598	0.668	0.627	0.644
CART	0.943	<u>0.608</u>	0.559	<u>0.497</u>	0.975	<u>0.780</u>	<u>0.813</u>	0.955	0.766
R²M (ours)	<u>0.930</u>	0.594	0.709	0.739	<u>0.964</u>	0.823	0.919	0.995	0.834

► *Training-free model merging fits deepfake detection.* Even plain *Weight Averaging* achieves a strong FF mean AUC of 0.982, essentially matching the jointly trained *All-in-one* model (0.979). This indicates substantial cross-task parameter sharing in this domain and suggests that weight-space merging is a natural and effective paradigm for deepfake detection.

► *CART and R²M retain specialists almost perfectly on FF (saturation).* Both *CART* and *R²M* achieve near-saturated performance on FF (Protocol 1), reaching mean AUCs of 0.988 and 0.996, respectively. Category-wise values closely track the specialists (e.g., FS: 0.976/0.992 vs 0.995; FR: 0.994/0.999 vs 0.999; EFS: 0.995/0.997 vs 0.999). These small differences (on the order of 10^{-2}) indicate that the in-domain FF setting is nearly saturated across merging methods.

► *Cross-domain seen transfer (CDF; Protocol 2).* When transferring from FF++ to Celeb-DF, performance differences become larger. Specialists can fail catastrophically under this setting. For example, the FR specialist achieves an AUC of only 0.099 on EFS under CDF, indicating that decision boundaries learned for one manipulation category may not transfer across domain and manipulation shifts. Training-free merging methods mitigate such failures. Among them, *CART* achieves the highest CDF mean AUC (0.877), while *R²M* remains competitive with a mean of 0.852. Notably, *R²M* achieves the best result on the FR column (0.932), suggesting that controlling real-sensitive modes while preserving specialist residual directions helps maintain robustness for manipulation types.

Table 3: Effect of spectral ranks (r_0, r_A) and calibration. We report mean AUCs on Protocol 1 (FF), Protocol 2 (CDF), and Protocol 3 (Unseen), before (off) and after (cal) suite-level calibration. Δ denotes (cal - off). All = mean of FF/CDF/Unseen.

r_0	r_A	FF _{off}	FF _{cal}	Δ	CDF _{off}	CDF _{cal}	Δ	U _{off}	U _{cal}	Δ	All _{off}	All _{cal}	Δ
0	1	0.967	0.979	+0.012	0.485	0.810	+0.325	0.728	0.769	+0.041	0.727	0.853	+0.126
0	2	0.967	0.988	+0.021	0.637	0.845	+0.208	0.738	0.777	+0.039	0.781	0.870	+0.089
0	3	0.974	0.990	+0.016	0.659	0.870	+0.211	0.718	0.752	+0.034	0.783	0.871	+0.088
1	1	0.956	0.987	+0.031	0.538	0.838	+0.300	0.738	0.777	+0.039	0.744	0.868	+0.124
1	2	0.989	0.989	+0.000	0.695	0.842	+0.147	0.786	0.822	+0.036	0.823	0.885	+0.062
1	3	0.990	0.990	+0.000	0.697	0.842	+0.145	0.781	0.814	+0.033	0.823	0.882	+0.059
2	1	0.975	0.980	+0.005	0.530	0.844	+0.314	0.730	0.764	+0.034	0.745	0.863	+0.118
2	2	0.991	0.991	+0.000	0.683	0.852	+0.169	0.777	0.816	+0.039	0.817	0.886	+0.069
2	3	0.996	0.996	+0.000	0.730	0.852	+0.122	0.798	0.834	+0.036	0.841	0.894	+0.053

Tab. 2 reports per-forgery AUC on *unseen* generators, with the macro mean.

► *Merging prevents catastrophic failures and improves macro performance.* The jointly trained *All-in-one* model collapses on several unseen generators (e.g., MidJourney AUC = 0.177). This suggests that a decision boundary learned for FR in FF++ does not transfer to EFS in CDF, indicating significant geometric misalignment across categories and domains. Similarly, the jointly trained All-in-one model, while competitive on FF++, drops to 0.366 on EFS under CDF, implying that collapsing heterogeneous forgeries into a single shared boundary can lead to suboptimal compromises under domain shift. In contrast, all training-free merging variants avoid such failures. For example, *Weight Averaging* achieves 0.613 and *CART* 0.559 on MidJourney. This shows that merging preserves diverse specialist cues and improves robustness under distribution shift.

► *R²M delivers the strongest overall zero-shot transfer.* Among training-free methods, **R²M** achieves the best macro AUC on unseen generators (0.834), outperforming CART (0.766), Weight Averaging (0.724), and other baselines. R²M achieves the highest performance on multiple generators, including MidJourney (0.709), WhichIsReal (0.739), StarGAN v2 (0.823), StyleCLIP (0.919), and CollabDiff (0.995). Overall, these results indicate that preserving manipulation-specific residual directions while controlling real-sensitive modes leads to stronger zero-shot generalization across diverse unseen generators.

4.2 Ablation on Spectral Construction and Calibration

All results in this rank-grid analysis are obtained with a fixed residual scaling coefficient $\beta = 1$. Therefore, performance differences arise solely from the spectral ranks (r_0, r_A) and the presence or absence of calibration (P3), not from tuning the residual magnitude. Tab. 3 summarizes the mean AUCs across Protocols 1–3 for all (r_0, r_A) configurations, before and after calibration.

Effect of Spectral Ranks (r_0, r_A) Across the rank grid, increasing r_A consistently improves the raw (off) basis. For example, under $r_0 = 2$, the overall mean increases from 0.745 ($r_A = 1$) to 0.841 ($r_A = 3$). This indicates that a higher residual rank better captures the second-order structure of specialist task vectors, enabling more accurate routing.

In contrast, setting $r_0 = 0$ (i.e., no Real-mode projection) leads to unstable cross-domain behavior. While FF performance may remain competitive, CDF performance degrades severely in the off setting (e.g., 0.485 for $r_0 = 0, r_A = 1$). Using $r_0 \geq 1$ consistently improves cross-domain robustness, supporting the role of Stage 1 as approximate Real-whitening.

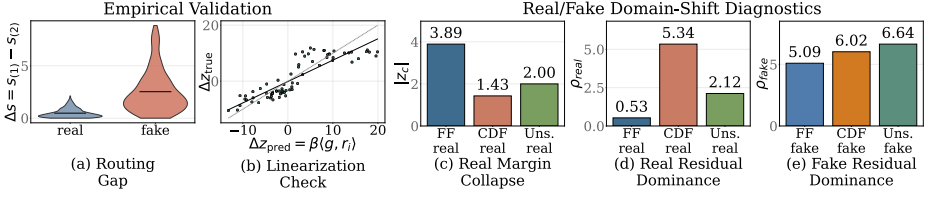


Fig. 3: Empirical validation of the proposed real-fake decomposition. (a) Routing confidence gap $\Delta s = s_{(1)} - s_{(2)}$ where $s_i(x) = |\langle g(x), r_i \rangle|$ is the routing score for residual branch i and $s_{(1)} \geq s_{(2)}$ are the top-2 scores. (b) Linearization check comparing predicted logit shifts $\Delta z_{\text{pred}} = \beta \langle g(x), r_i \rangle$ with actual shifts Δz_{true} . (c) Common margin magnitude $|z_c|$ on real samples across domains. (d) Residual-to-common ratio ρ on real samples. (e) Residual-to-common ratio ρ on fake samples.

Interaction Between r_0 and Calibration (P3). Calibration (P3) plays a complementary role to the spectral construction stages (P1–P2), and its effect depends strongly on the quality of the residual basis.

First, when the residual subspace is weak—*e.g.* small r_A and no Real-mode projection ($r_0 = 0$)—calibration yields substantial improvements, especially under domain shift. For instance, with $(r_0, r_A) = (0, 1)$, calibration improves the CDF mean by $+0.325$. In this regime, the residual directions are both low-rank and sensitive to Real-dominant modes, leading to unstable residual magnitudes. Calibration compensates for this mismatch by explicitly regulating the residual contribution relative to the shared margin. Second, as r_0 and r_A increase, the gap between off and calibrated performance tends to shrink, particularly in the in-domain FF setting. For the strongest basis $(r_0, r_A) = (2, 3)$, the FF gain from calibration is zero ($+0.000$). This suggests that stronger Real-mode suppression (P1) together with higher residual capacity (P2) already produce more stable residual magnitudes, reducing reliance on inference-time correction.

Nevertheless, calibration remains important under domain shift. Even with the strongest basis, CDF performance improves by $+0.122$ after calibration. This indicates that while improved spectral construction reduces magnitude instability in seen domains, cross-domain shifts still induce scale mismatch that is not fully captured offline. Therefore, P3 does not replace P1–P2; rather, it stabilizes their interaction, particularly in challenging transfer settings. Additional ablations on the residual scaling coefficient β are provided in Appendix B.3 showing that the overall trends remain consistent across scaling values.

4.3 Empirical Analysis of Real-Fake Decomposition

Routing selectivity. We score each residual branch by $s_i(x) = |\langle g(x), r_i \rangle|$, and route by the top score. Fig. 3(a) plots the confidence gap $\Delta s = s_{(1)} - s_{(2)}$ between the top-2 scores. Fake samples exhibit substantially larger gaps than real samples, indicating more selective residual activation for manipulated inputs.

Linearization validity. Fig. 3(b) compares the predicted logit shift $\Delta z_{\text{pred}} = \beta \langle g(x), r_i \rangle$ with the actual shift Δz_{true} obtained by applying the residual update. The strong correlation between the two quantities indicates that the gradient inner-product provides a reliable first-order surrogate for estimating the effect of residual updates, supporting the linearization assumption used for routing. To

better understand the behavior of the proposed decomposition, we analyze the statistics of the common margin and residual routing under domain shifts.

Real-domain mismatch. Fig. 3(c) shows the magnitude of the common margin $|z(\theta_c, x)|$ on real samples across domains. While the FF real distribution produces large margins, the margin on CDF real drops, indicating that the FF-derived real-sensitive subspace is less aligned with the CDF real distribution.

Residual dominance under domain shift. In Fig. 3(d), the residual-to-common ratio $\rho = |\Delta z|/|z_c|$ increases sharply on CDF real data. This suggests that the weakened common margin causes residual routing to dominate more frequently, and residual routing dominates when the common margin collapses.

Residual selectivity for unseen generators. In contrast, Fig. 3(e) shows that unseen fake samples increase the relative importance of the residual term. This indicates that manipulation-specific residual directions remain informative for unseen fake families, supporting improved generalization in the unseen setting.

4.4 Merging specialists with heterogeneous real domains

Our analysis so far used DF40, where all real images come from FF, raising the question of whether the observed “shared Real” structure persists under more heterogeneous real distributions. To test this, we train specialists using two real datasets (**FF** and **CDF**) while keeping the same forgery families, yielding six specialists in total. We merge them using CART or R^2M and evaluate AUC across the six domains. The results are summarized

Table 4: Merging six specialists with heterogeneous real distributions. AUC on six domains for the individual specialists (FS-ff, FR-ff, EFS-ff, FS-cdf, FR-cdf, EFS-cdf), and for the merged models obtained by CART and R^2M .

Model	FS-ff	FR-ff	EFS-ff	FS-cdf	FR-cdf	EFS-cdf
FS-ff	0.995	0.912	0.766	0.959	0.696	0.621
FR-ff	0.923	0.999	0.742	0.503	0.923	0.106
EFS-ff	0.676	0.814	0.999	0.635	0.608	0.988
FS-cdf	0.754	0.647	0.616	0.999	0.989	0.982
FR-cdf	0.557	0.849	0.532	0.991	0.999	0.929
EFS-cdf	0.329	0.244	0.923	0.955	0.961	0.999
CART	0.867	0.957	0.954	0.998	0.994	0.996
R^2M	0.971	0.989	0.992	0.996	0.994	0.981

in Tab. 4. Despite the increased real heterogeneity, R^2M consistently achieves higher AUC than CART on the harder FF++ domains while matching CART on the saturated CDF domains. This suggests that the SVD-based Real core in R^2M maintains a stabilizing shared subspace across multiple real datasets and degrades more gracefully as real diversity increases. Additional scaling experiments and training details are provided in the Appendix B.5 and B.1.

5 Conclusion

We study deepfake detection as a testbed for model merging, where specialists share a binary objective but capture generator-specific artifacts. R^2M links parameter-space composition to logit-space behavior, enabling efficient inference and interpretable analysis. Across in-domain and unseen settings, generator-specific residuals provide selective gains, while unconditional residual injection introduces cross-family trade-offs. Routing confidence and first-order logit prediction explain when residual specialization helps or fails. However, real-domain mismatch remains a key limitation, particularly under CDF-like shifts where margin collapse and residual dominance degrade ranking stability. R^2M combines offline spectral organization with lightweight routing to enable scalable and interpretable detector updates under distribution shift.

References

1. Heygen. <https://www.heygen.com/>, accessed 2025-09-25
2. Midjourney. <https://www.midjourney.com/>, accessed 2025-09-25
3. Which face is real? <https://www.whichfaceisreal.com/>, accessed 2025-09-25
4. Faceswap. <https://github.com/deepfakes/faceswap> (2019)
5. Chen, R., Chen, X., Ni, B., Ge, Y.: Simswap: An efficient framework for high fidelity face swapping. In: ACM MM. pp. 2003–2011 (2020)
6. Choi, J., Kim, D., Lee, C., Hong, S.: Revisiting weight averaging for model merging. arXiv preprint arXiv:2412.12153 (2024)
7. Choi, J., Kim, T., Jeong, Y., Baek, S., Choi, J.: Exploiting style latent flows for generalizing deepfake video detection. In: CVPR. pp. 1133–1143 (2024)
8. Choi, Y., Choi, M., Kim, M., Ha, J.W., Kim, S., Choo, J.: Stargan: Unified generative adversarial networks for multi-domain image-to-image translation. In: CVPR. pp. 8789–8797 (2018)
9. Choi, Y., Uh, Y., Yoo, J., Ha, J.W.: Stargan v2: Diverse image synthesis for multiple domains. In: CVPR. pp. 8188–8197 (2020)
10. Cui, X., Li, Y., Luo, A., Zhou, J., Dong, J.: Forensics adapter: Adapting clip for generalizable face forgery detection. In: CVPR. pp. 19207–19217 (2025)
11. Fedus, W., Zoph, B., Shazeer, N.: Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research* **23**(120), 1–39 (2022)
12. Guo, J., Zhang, D., Liu, X., Zhong, Z., Zhang, Y., Wan, P., Zhang, D.: Liveportrait: Efficient portrait animation with stitching and retargeting control. arXiv preprint arXiv:2407.03168 (2024)
13. Huang, Z., Chan, K.C.K., Jiang, Y., Liu, Z.: Collaborative diffusion for multi-modal face generation and editing. In: CVPR. pp. 6080–6090 (2023)
14. Ilharco, G., Ribeiro, M.T., Wortsman, M., Gururangan, S., Schmidt, L., Hajishirzi, H., Farhadi, A.: Editing models with task arithmetic. arXiv preprint arXiv:2212.04089 (2022)
15. Izmailov, P., Podoprikin, D., Garipov, T., Vetrov, D., Wilson, A.G.: Averaging weights leads to wider optima and better generalization. arXiv preprint arXiv:1803.05407 (2018)
16. Jeong, Y., Kim, D., Ro, Y., Choi, J.: Frepgan: robust deepfake detection using frequency-level perturbations. In: AAAI. pp. 1060–1068 (2022)
17. Li, J., Zhang, J., Bai, X., Zheng, J., Ning, X., Zhou, J., Gu, L.: Talkinggaussian: Structure-persistent 3d talking head synthesis via gaussian splatting. In: ECCV. pp. 127–145 (2024)
18. Li, L., Bao, J., Yang, H., Chen, D., Wen, F.: Faceshifter: Towards high fidelity and occlusion aware face swapping. arXiv preprint arXiv:1912.13457 (2019)
19. Li, L., Bao, J., Yang, H., Chen, D., Wen, F.: Advancing high fidelity identity swapping for forgery detection. In: CVPR. pp. 5074–5083 (2020)
20. Li, L., Bao, J., Zhang, T., Yang, H., Chen, D., Wen, F., Guo, B.: Face x-ray for more general face forgery detection. In: CVPR. pp. 5001–5010 (2020)
21. Li, Y., Yang, X., Sun, P., Qi, H., Lyu, S.: Celeb-df: A large-scale challenging dataset for deepfake forensics. In: CVPR. pp. 3207–3216 (2020)
22. Lin, Y., Song, W., Li, B., Li, Y., Ni, J., Chen, H., Li, Q.: Fake it till you make it: Curricular dynamic forgery augmentations towards general deepfake detection. In: ECCV. pp. 104–122 (2024)

23. Liu, K., Perov, I., Gao, D., Chervoniyy, N., Zhou, W., Zhang, W.: Deepfacelab: Integrated, flexible and extensible face-swapping framework. *Pattern Recognition* **141**, 109628 (2023)
24. Matena, M.S., Raffel, C.A.: In: Merging models with fisher-weighted averaging. pp. 17703–17716 (2022)
25. Mukhopadhyay, S., Suri, S., Gadde, R.T., Shrivastava, A.: Diff2lip: Audio conditioned diffusion models for lip-synchronization. In: WACV. pp. 5292–5302 (2024)
26. Nirkin, Y., Keller, Y., Hassner, T.: Fsgan: Subject agnostic face swapping and reenactment. In: ICCV. pp. 7184–7193 (2019)
27. Park, J., Owens, A.: Community forensics: Using thousands of generators to train fake image detectors. In: CVPR. pp. 8245–8257 (2025)
28. Patashnik, O., Wu, Z., Shechtman, E., Cohen-Or, D., Lischinski, D.: Styleclip: Text-driven manipulation of stylegan imagery. In: ICCV. pp. 2085–2094 (2021)
29. Prajwal, K., Mukhopadhyay, R., Namboodiri, V.P., Jawahar, C.: A lip sync expert is all you need for speech to lip generation in the wild. In: ACM MM. pp. 484–492 (2020)
30. Radford, A., Kim, J.W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., et al.: Learning transferable visual models from natural language supervision. In: ICML. pp. 8748–8763 (2021)
31. Richardson, E., Alaluf, Y., Patashnik, O., Nitzan, Y., Azar, Y., Shapiro, S., Cohen-Or, D.: Encoding in style: a stylegan encoder for image-to-image translation. In: CVPR. pp. 2287–2296 (2021)
32. Rombach, R., Blattmann, A., Lorenz, D., Esser, P., Ommer, B.: High-resolution image synthesis with latent diffusion models. In: CVPR. pp. 10684–10695 (2022)
33. Rossler, A., Cozzolino, D., Verdoliva, L., Riess, C., Thies, J., Nießner, M.: Faceforensics++: Learning to detect manipulated facial images. In: ICCV. pp. 1–11 (2019)
34. Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q., Hinton, G., Dean, J.: Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. *arXiv preprint arXiv:1701.06538* (2017)
35. Shiohara, K., Yamasaki, T.: Detecting deepfakes with self-blended images. In: CVPR. pp. 18720–18729 (2022)
36. Standley, T., Zamir, A., Chen, D., Guibas, L., Malik, J., Savarese, S.: Which tasks should be learned together in multi-task learning? In: ICML. pp. 9120–9132 (2020)
37. Sun, K., Chen, S., Yao, T., Zhou, Z., Ji, J., Sun, X., Lin, C.W., Ji, R.: Towards general visual-linguistic face forgery detection. In: CVPR. pp. 19576–19586 (2025)
38. Tan, C., Zhao, Y., Wei, S., Gu, G., Liu, P., Wei, Y.: Frequency-aware deepfake detection: Improving generalizability through frequency space domain learning. In: AAAI. pp. 5052–5060 (2024)
39. Thies, J., Zollhöfer, M., Nießner, M.: Deferred neural rendering: Image synthesis using neural textures. *Acm Transactions on Graphics* **38**(4), 1–12 (2019)
40. Tolosana, R., Vera-Rodriguez, R., Fierrez, J., Morales, A., Ortega-Garcia, J.: Deepfakes and beyond: A survey of face manipulation and fake detection. *Information Fusion* **64**, 131–148 (2020)
41. Wortsman, M., Ilharco, G., Gadre, S.Y., Roelofs, R., Gontijo-Lopes, R., Morcos, A.S., Namkoong, H., Farhadi, A., Carmon, Y., Kornblith, S., et al.: Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In: ICML. pp. 23965–23998 (2022)
42. Yadav, P., Tam, D., Choshen, L., Raffel, C.A., Bansal, M.: In: Ties-merging: Resolving interference when merging models. pp. 7093–7115 (2023)

43. Yan, Z., Wang, J., Wang, Z., Jin, P., Zhang, K.Y., Chen, S., Yao, T., Ding, S., Wu, B., Yuan, L.: Effort: Efficient orthogonal modeling for generalizable ai-generated image detection. arXiv preprint arXiv:2411.15633 **2** (2024)
44. Yan, Z., Yao, T., Chen, S., Zhao, Y., Fu, X., Zhu, J., Luo, D., Wang, C., Ding, S., Wu, Y., et al.: Df40: Toward next-generation deepfake detection. Advances in Neural Information Processing Systems **37**, 29387–29434 (2024)
45. Yan, Z., Zhang, Y., Fan, Y., Wu, B.: Ucf: Uncovering common features for generalizable deepfake detection. In: ICCV. pp. 22412–22423 (2023)
46. Yang, E., Wang, Z., Shen, L., Liu, S., Guo, G., Wang, X., Tao, D.: Adamerging: Adaptive model merging for multi-task learning. arXiv preprint arXiv:2310.02575 (2023)
47. Yu, T., Kumar, S., Gupta, A., Levine, S., Hausman, K., Finn, C.: Gradient surgery for multi-task learning. In: NIPS. pp. 5824–5836 (2020)
48. Zhou, J., Li, Y., Wu, B., Li, B., Dong, J., et al.: Freqblender: Enhancing deepfake detection by blending frequency knowledge. In: NeurIPS. pp. 44965–44988 (2024)
49. Zhou, Y., Lei, T., Liu, H., Du, N., Huang, Y., Zhao, V., Dai, A.M., Le, Q.V., Laudon, J., et al.: Mixture-of-experts with expert choice routing. Advances in Neural Information Processing Systems **35**, 7103–7114 (2022)
50. Zhu, X., Guan, Y., Liang, D., Chen, Y., Liu, Y., Bai, X.: Moe jetpack: From dense checkpoints to adaptive mixture of experts for vision tasks. Advances in Neural Information Processing Systems **37**, 12094–12118 (2024)